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APTITUDE QUESTION BANK

Programming & Technical Aptitude

Placement preparation for IT company hiring tests

Where the Syllabus Ends and Reality Begins

150	Q1 – Q50	Q51 – Q100	Q101 – Q150
Multiple-choice questions	Beginner	Intermediate	Advanced

About this section

Programming and Technical Aptitude questions check whether you can read code and reason about it: predicting output, tracing loops, understanding data structures, algorithms, time complexity and object-oriented concepts. Snippets here use Python, C and Java, which are the most common languages in campus tests. Trace the code line by line on paper, track the value of each variable, and watch for off-by-one errors.

How to use this question bank

- Part A has all 150 questions, grouped by difficulty. Attempt them on your own first, with a timer if you can (about 1 minute per question).
- Part B is the answer key. Check your answers only after finishing a full level.
- Part C explains how to solve every question in simple steps. Read the solution even for the ones you got right — it often shows a faster method.
- Mark the questions you got wrong and attempt them again after two or three days.

PART A

Questions

BEGINNER Questions 1 – 50

Q1. What is the output of this Python code?

```
# Python
print(5 + 3)
```

A) Error
B) 8
C) 5
D) 35

Q2. What is the output?

```
# Python
print('5' + '3')
```

A) 53
B) 8
C) 5 3
D) Error

Q3. Which data structure works on the principle 'Last In, First Out'?

A) Queue
B) Tree
C) Stack
D) Array

Q4. Which data structure works on 'First In, First Out'?

A) Stack
B) Heap
C) Graph
D) Queue

Q5. What is the output?

```
# Python
print(7 // 2)
```

A) 3
B) 2
C) 3.5
D) 4

Q6. What is the output?

```
# Python
print(7 % 2)
```

A) 3
B) 1
C) 2
D) 0

Q7. What will this loop print?

```
# Python
for i in range(3):
    print(i, end=' ')
```

A) 1 2
B) 1 2 3
C) 0 1 2
D) 0 1 2 3

Q8. Which keyword is used to define a function in Python?

A) fun
B) define
C) function
D) def

Q9. What is the index of the first element in a Python list or C array?

A) 1
B) 0
C) −1
D) Depends on the size

Q10. What is the output?

```
# Python
x = 5
x = x * 2
print(x)
```

A) 10
B) 15
C) Error
D) 5

Q11. Which of these is a valid variable name in most languages?

A) total-marks
B) total marks
C) 2marks
D) total_marks

Q12. What does the following C code print?

```
// C
#include <stdio.h>
int main() {
    printf("Hello");
    return 0;
}
```

A) Nothing
B) "Hello"
C) Hello
D) hello

Q13. What is the output?

```
# Python
print(10 > 5)
```

A) True
B) Error
C) 1
D) False

Q14. What is an algorithm?

A) A programming language
B) A type of computer
C) A step-by-step procedure to solve a problem
D) A software bug

Q15. What is the output?

```
# Python
nums = [10, 20, 30]
print(len(nums))
```

A) Error
B) 3
C) 2
D) 4

Q16. What is the output?

```
# Python
nums = [10, 20, 30]
print(nums[1])
```

A) Error
B) 30
C) 10
D) 20

Q17. What is the output?

```
# Python
nums = [10, 20, 30]
print(nums[-1])
```

A) 30
B) 10
C) 20
D) Error

Q18. Which symbol is used for single-line comments in Python?

A) --
B) /*
C) //
D) #

Q19. What is the output?

```
# Python
n = 8
if n % 2 == 0:
    print('Even')
else:
    print('Odd')
```

A) Error
B) Odd
C) Even
D) Nothing

Q20. What is the output?

```
# Python
print('hello'.upper())
```

A) Error
B) HELLO
C) hello
D) Hello

Q21. Which of these is NOT a loop in C?

A) do-while
B) for
C) while
D) repeat-until

Q22. What is the output?

```
# Python
total = 0
for i in range(1, 4):
    total += i
print(total)
```

A) Error
B) 5
C) 6
D) 4

Q23. What does the 'return' statement do in a function?

A) Sends a value back to the caller and ends the function
B) Deletes the function
C) Repeats the function
D) Prints a value

Q24. What is the output?

```
# Python
print(3 ** 2)
```

A) 6
B) 9
C) Error
D) 8

Q25. What is a 'bug' in programming?

A) A type of loop
B) An error or flaw in a program
C) A fast program
D) A computer virus

Q26. What is the output?

```
# Python
print(type(5))
```

A) int
B) <class 'float'>
C) <class 'int'>
D) <class 'str'>

Q27. Which data type is used to store True or False?

A) Boolean
B) Integer
C) String
D) Float

Q28. What is the output?

```
# Python
for i in range(3):
    print('Hi', end=' ')
```

A) HiHiHi3
B) Error
C) Hi
D) Hi Hi Hi

Q29. What is the value of x after this C code?

```
// C
int x = 10;
x++;
```

A) 11
B) 12
C) 10
D) 9

Q30. Which operator checks equality in Python, C and Java?

A) !=
B) ===
C) ==
D) =

Q31. What is the output?

```
# Python
a = [1, 2, 3]
a.append(4)
print(a)
```

A) [1, 2, 3]
B) Error
C) [4, 1, 2, 3]
D) [1, 2, 3, 4]

Q32. What is the output?

```
# Python
s = 'Python'
print(s)
```

A) python
B) Python
C) Error
D) P

Q33. What is the output?

```
# Python
s = 'Python'
print(s[0])
```

A) P
B) y
C) Error
D) n

Q34. Which of these is used to take input from the user in Python?

A) get()
B) input()
C) scan()
D) read()

Q35. What does 'OOP' stand for?

A) Object Ordered Programming
B) Open Online Programming
C) Only One Program
D) Object-Oriented Programming

Q36. In OOP, what is a 'class'?

A) A loop
B) A single object
C) A blueprint for creating objects
D) A variable

Q37. What is the output?

```
# Python
print(max(2, 5, 3))
```

A) Error
B) 5
C) 3
D) 2

Q38. What is the output?

```
# Python
print(5 / 2)
```

A) 2.5
B) 3
C) 2
D) Error

Q39. What will this code print?

```
# Python
score = 75
if score >= 50:
    print('Good')
else:
    print('Bad')
```

A) Error
B) Nothing
C) Good
D) Bad

Q40. What is the output?

```
# Python
print('abc' * 2)
```

A) abc2
B) abc abc
C) Error
D) abcabc

Q41. What is 'debugging'?

A) Deleting a program
B) Installing software
C) Writing new code
D) Finding and fixing errors in code

Q42. Which of these is a compiled language (traditionally)?

A) HTML
B) C
C) Markdown
D) CSS

Q43. What is the output?

```
# Python
print(min([3, 7, 5]))
```

A) 1
B) 0
C) 3
D) Error

Q44. What is the output?

```
# Python
def add(a, b):
    return a + b
print(add(10, 5))
```

A) 15
B) 10
C) 5
D) Error

Q45. What is a 'variable'?

A) A function
B) A type of loop
C) A fixed number
D) A named storage location for data

Q46. What is the output?

```
# Python
print(True and False)
```

A) Error
B) True
C) False
D) 0

Q47. What is the output?

```
# Python
print(True or False)
```

A) False
B) True
C) 1
D) Error

Q48. What is the output?

```
# Python
a = [1, 2, 3, 4]
print(a[1:3])
```

A) [2, 3]
B) [2, 3, 4]
C) [1, 2]
D) [1, 2, 3]

Q49. Which of these sorting methods repeatedly swaps neighbouring elements that are in the wrong order?

A) Bubble sort
B) Quick sort
C) Merge sort
D) Binary search

Q50. What is the output?

```
# Python
i = 0
while i < 4:
    i += 1
print(i)
```

A) Error
B) 5
C) 3
D) 4

INTERMEDIATE Questions 51 – 100

Q51. What is the output?

```
# Python
a = [1, 2, 3]
b = a
b.append(4)
print(a)
```

A) [1, 2, 3]
B) Error
C) [1, 2, 3, 4]
D) [4]

Q52. What is the time complexity of binary search on a sorted array of n elements?

A) O(1)
B) O(log n)
C) O(n)
D) $O(n^2)$

Q53. What is the output?

```
# Python
print('hello'[::-1])
```

A) olleh
B) Error
C) h
D) hello

Q54. What is the output?

```
# Python
def fact(n):
    if n <= 1:
        return 1
    return n * fact(n - 1)
print(fact(5))
```

A) 720
B) 120
C) 24
D) Error

Q55. What is the worst-case time complexity of bubble sort?

A) O(n log n)
B) O(n)
C) $O(n^2)$
D) O(log n)

Q56. What is the output of this C code?

```
// C
void swap(int x, int y) {
    int t = x; x = y; y = t;
}
int main() {
    int a = 10, b = 20;
    swap(a, b);
    printf("%d %d", a, b);
}
```

A) 10 10
B) 20 20
C) 20 10
D) 10 20

Q57. Which OOP principle hides internal details and exposes only what is necessary?

A) Encapsulation
B) Inheritance
C) Polymorphism
D) Recursion

Q58. Which OOP principle lets a child class reuse the properties of a parent class?

A) Overloading
B) Encapsulation
C) Inheritance
D) Abstraction

Q59. What is the output?

```
# Python
d = {}
d['a'] = 1
d['b'] = 2
print(d)
```

A) {1, 2}
B) ['a', 'b']
C) Error
D) {'a': 1, 'b': 2}

Q60. What is the output?

```
# Python
s = {1, 2, 2, 3}
print(len(s))
```

A) 2
B) 3
C) Error
D) 4

Q61. What is the output?

```
# Python
for i in range(1, 4):
    print(i * i, end=' ')
```

A) 2 4 6
B) 1 8 27
C) 1 2 3
D) 1 4 9

Q62. Which data structure is best for checking whether brackets in an expression are balanced?

A) Stack
B) Linked list
C) Queue
D) Hash table

Q63. What is the output?

```
# Python
print([x * 2 for x in range(5)])
```

A) [2, 4, 6, 8]
B) [0, 2, 4, 6, 8, 10]
C) [0, 2, 4, 6, 8]
D) [0, 1, 2, 3, 4]

Q64. What is the output of this Java code?

```
// Java
int a = 10, b = 5;
System.out.println(a + b);
```

A) 510
B) 15
C) 105
D) Error

Q65. What is the output of this Java code?

```
// Java
System.out.println("Sum: " + 10 + 5);
```

A) Sum: 105
B) Error
C) Sum: 10 5
D) Sum: 15

Q66. Which of these has O(1) average time for lookup by key?

A) Sorted array
B) Binary tree (unbalanced)
C) Linked list
D) Hash table

Q67. What is the output?

```
# Python
print('level'.count('l'))
```

A) Error
B) 2
C) 1
D) 3

Q68. What is the output?

```
# Python
print('hello world'.find(' '))
```

A) Error
B) 6
C) 5
D) 4

Q69. What does 'polymorphism' mean in OOP?

A) The same method name behaving differently for different objects
B) Creating many classes
C) Hiding data
D) Using many variables

Q70. What is the output?

```
# Python
count = 0
for i in range(0, 9, 3):
    count += 1
print(count)
```

A) 6
B) 9
C) 3
D) 0

Q71. What is the output of this C code?

```
// C
int x = 5;
int y = x++;
printf("%d", y);
```

A) Undefined
B) 5
C) 4
D) 6

Q72. What is the output of this C code?

```
// C
int x = 5;
int y = ++x;
printf("%d", y);
```

A) Undefined
B) 5
C) 7
D) 6

Q73. Which traversal of a binary search tree gives the values in sorted (ascending) order?

A) Post-order
B) Level-order
C) In-order
D) Pre-order

Q74. What is the output?

```
# Python
a = [1, 2, 3]
a.reverse()
print(a)
```

A) Error
B) [1, 2, 3]
C) None
D) [3, 2, 1]

Q75. What is the output?

```
# Python
a = [1, 2, 3]
print(a.reverse())
```

A) None
B) [1, 2, 3]
C) [3, 2, 1]
D) Error

Q76. What is a linked list?

A) An array of fixed size
B) A sequence of nodes where each node points to the next
C) A type of loop
D) A sorted table

Q77. What is the output?

```
# Python
x = 10
def change():
    x = 5
change()
print(x)
```

A) 10
B) 0
C) Error
D) 5

Q78. What is the output?

```
# Python
for i in range(1, 6):
    if i % 3 == 0:
        print('Fizz', end=' ')
    elif i % 5 == 0:
        print('Buzz', end=' ')
    else:
        print(i, end=' ')
```

A) 1 2 3 4 5
B) Fizz Buzz
C) 1 2 Fizz 4 Buzz
D) 1 2 Fizz 4 5

Q79. What is the output?

```
# Python
print(sum([1, 2, 3, 4, 5]))
```

A) 5
B) Error
C) 10
D) 15

Q80. What is the output?

```
# Python
print(list('abc'))
```

A) 'abc'
B) ['a', 'b', 'c']
C) Error
D) ['abc']

Q81. What is the output?

```
# Python
print('red,green,blue'.split(','))
```

A) Error
B) red green blue
C) ['red,green,blue']
D) ['red', 'green', 'blue']

Q82. Which sorting algorithm uses 'divide and conquer' and has O(n log n) time in all cases?

- **A)** Selection sort
- **B)** Merge sort
- **C)** Insertion sort
- **D)** Bubble sort

Q83. What is the output?

```
# Python
for i in range(5):
    if i == 2:
        break
print(i)
```

- **A)** 3
- **B)** 0
- **C)** 2
- **D)** 1

Q84. What is the output?

```
# Python
for i in range(5):
    if i == 2:
        continue
    print(i, end=' ')
```

- **A)** 0 1 3 4
- **B)** 0 1
- **C)** 0 1 2 3 4
- **D)** 2

Q85. What is the output?

```
# Python
t = (1, 2, 3)
print(t)
```

- **A)** {1, 2, 3}
- **B)** Error
- **C)** (1, 2, 3)
- **D)** [1, 2, 3]

Q86. What happens when you run this Python code?

```
# Python
t = (1, 2, 3)
t[0] = 9
```

- **A)** Nothing
- **B)** TypeError — tuples cannot be changed
- **C)** It prints (9, 2, 3)
- **D)** It prints 9

Q87. Which keyword in Java is used to create a new object?

- **A)** new
- **B)** create
- **C)** make
- **D)** object

Q88. What is the output?

```
# Python
print(3 in [1, 2, 3])
```

- **A)** False
- **B)** None
- **C)** Error
- **D)** True

Q89. What is the space complexity of storing an n × n matrix?

- **A)** O(n)
- **B)** O(log n)
- **C)** $O(n^2)$
- **D)** O(1)

Q90. What is the output?

```
# Python
def double(x):
    return x * 2
print(double(double(1)))
```

- **A)** 16
- **B)** 2
- **C)** 8
- **D)** 4

Q91. What is the output?

```
# Python
print(int('3') * int('4'))
```

- **A)** 34
- **B)** 12
- **C)** 7
- **D)** Error

Q92. What is a 'constructor' in OOP?

- **A)** A special method that runs when an object is created
- **B)** A loop inside a class
- **C)** A global variable
- **D)** A method that deletes objects

Q93. What is the output?

```
# Python
name = 'Asha'
print(f'Hello, {name}')
```

A) Hello, name
B) Error
C) Hello, Asha
D) Hello,

Q94. What is the output?

```
# Python
n = 12345
count = 0
while n > 0:
    n //= 10
    count += 1
print(count)
```

A) 5
B) 4
C) 6
D) Error

Q95. What is the output?

```
# Python
print(sorted([3, 1, 4, 5, 2]))
```

A) [5, 4, 3, 2, 1]
B) [1, 2, 3, 4, 5]
C) None
D) [3, 1, 4, 5, 2]

Q96. Which of these correctly describes a queue implemented with 'enqueue' and 'dequeue'?

A) Enqueue removes from the front
B) Enqueue adds at the rear; dequeue removes from the front
C) Dequeue adds at the rear
D) Both work at the same end

Q97. What is the output?

```
# Python
def f(x=10):
    return x
print(f())
```

A) Error
B) 1
C) 0
D) 10

Q98. What does the 'break' statement do in a loop?

- **A)** Exits the loop immediately
- **B)** Pauses the program
- **C)** Restarts the loop
- **D)** Skips to the next iteration

Q99. What is the output?

```
# Python
print('ABC'.lower())
```

- **A)** ABC
- **B)** abc
- **C)** Abc
- **D)** aBC

Q100. What is the output?

```
# Python
print([10, 20, 30, 40].index(40))
```

- **A)** 2
- **B)** 1
- **C)** 3
- **D)** Error

ADVANCED Questions 101 – 150

Q101. What is the output?

```
# Python
def add(x, lst=[]):
    lst.append(x)
    return lst
print(add(1), end=' ')
print(add(1))
```

A) Error
B) [1] [1, 1]
C) [1] [1]
D) [1, 1] [1, 1]

Q102. What is the output?

```
# Python
def fib(n):
    if n < 2:
        return n
    return fib(n - 1) + fib(n - 2)
print(fib(6))
```

A) 13
B) 5
C) 21
D) 8

Q103. What is the time complexity of the naive recursive Fibonacci function?

A) $O(2^n)$
B) O(log n)
C) $O(n^2)$
D) O(n)

Q104. What is the output?

```
# Python
grid = [[0] * 2] * 2
grid[0][0] = 1
print(grid)
```

A) [[1, 0], [0, 0]]
B) [[0, 0], [0, 0]]
C) [[1, 0], [1, 0]]
D) Error

Q105. What is the output of this C code?

```c
// C
void add5(int *p) { *p = *p + 5; }
int main() {
    int a = 10;
    add5(&a);
    printf("%d", a);
}
```

A) 10
B) 5
C) Garbage value
D) 15

Q106. What is the worst-case time complexity of quicksort?

A) $O(n^2)$
B) O(n log n)
C) O(n)
D) O(log n)

Q107. What is the output?

```python
# Python
stack = []
stack.append(1)
stack.append(2)
stack.append(3)
print(stack.pop())
```

A) 1
B) 3
C) 2
D) 6

Q108. What is the output?

```python
# Python
from collections import deque
q = deque()
q.append(1)
q.append(2)
q.append(3)
print(q.popleft())
```

A) 3
B) Error
C) 1
D) 2

Q109. Which data structure is used for Breadth-First Search (BFS) of a graph?

A) Hash set only
B) Heap
C) Queue
D) Stack

Q110. What is the output?

```
# Python
counts = {}
for ch in 'aba':
    counts[ch] = counts.get(ch, 0) + 1
print(counts)
```

A) {'a': 2, 'b': 1}
B) Error
C) {'a': 1, 'b': 1}
D) {'b': 1, 'a': 2}

Q111. What is the output?

```
# Python
c = 0
for i in range(2):
    for j in range(3):
        c += 1
print(c)
```

A) 3
B) 0
C) 9
D) 6

Q112. What is the time complexity of this code?

```
# Python
for i in range(n):
    for j in range(n):
        print(i, j)
```

A) O(n log n)
B) $O(n^2)$
C) O(n)
D) O(log n)

Q113. What is the time complexity of this code?

```
# Python
i = 1
while i < n:
    i = i * 2
```

A) O(n)
B) O(log n)
C) O(1)
D) $O(n^2)$

Q114. What is the output of this Java code?

```
// Java
String a = new String("hi");
String b = new String("hi");
System.out.println((a == b) + " " + a.equals(b));
```

A) false true
B) false false
C) true true
D) true false

Q115. What is the output?

```
# Python
nums = [1, 2, 3, 4, 5]
print(list(filter(lambda x: x % 2 == 0, nums)))
```

A) [1, 2, 3, 4, 5]
B) [4, 16]
C) [1, 3, 5]
D) [2, 4]

Q116. What is the output?

```
# Python
print(list(map(lambda x: x ** 2, [1, 2, 3])))
```

A) Error
B) [2, 4, 6]
C) [1, 4, 9]
D) [1, 2, 3]

Q117. What does this function return for n = 1234?

```
# Python
def f(n):
    s = 0
    while n > 0:
        s += n % 10
        n //= 10
    return s
print(f(1234))
```

A) 10
B) 4
C) 24
D) 1234

Q118. What is the output?

```
# Python
def is_pal(s):
    return s == s[::-1]
print(is_pal('madam'))
```

A) Error
B) True
C) None
D) False

Q119. Which of these sorting algorithms is stable and works in O(n log n) time?

A) Selection sort
B) Heap sort
C) Merge sort
D) Quick sort (standard)

Q120. What is the output?

```
# Python
class Counter:
    count = 0
    def __init__(self):
        Counter.count += 1
a = Counter()
b = Counter()
print(Counter.count)
```

A) 1
B) Error
C) 3
D) 2

Q121. What is the output?

```
# Python
class Animal:
    def speak(self):
        return 'Animal sound'
class Dog(Animal):
    def speak(self):
        return 'Woof'
print(Dog().speak())
```

A) None
B) Animal sound
C) Woof
D) Error

Q122. What is the output?

```
# Python
try:
    print(10 / 0)
except ZeroDivisionError:
    print('Error: division by zero')
```

A) 0
B) Infinity
C) None
D) Error: division by zero

Q123. What is the output?

```
# Python
try:
    print('A', end=' ')
except Exception:
    print('B', end=' ')
finally:
    print('C')
```

A) A C
B) A
C) B C
D) A B C

Q124. What is the minimum number of comparisons needed in the worst case to find an item using binary search in a sorted list of 1,000 items?

A) 1000
B) 10
C) 100
D) 500

Q125. Which of these is an example of a 'greedy' algorithm?

A) Binary search
B) Merge sort
C) Bubble sort
D) Dijkstra's shortest path algorithm

Q126. What is the output?

```
# Python
a = [1, 2, 3]
b = a.copy()
b.append(4)
print(a, b)
```

A) [1, 2, 3, 4] [1, 2, 3, 4]
B) [1, 2, 3] [1, 2, 3]
C) [1, 2, 3] [1, 2, 3, 4]
D) Error

Q127. What is the output?

```
# Python
def gen():
    for i in range(3):
        yield i
for v in gen():
    print(v, end=' ')
```

A) 0 1 2
B) Nothing
C) 1 2 3
D) 0 1 2 3

Q128. What is the output?

```
# Python
from functools import reduce
print(reduce(lambda a, b: a + b, [1, 2, 3, 4, 5]))
```

A) Error
B) 15
C) 120
D) 6

Q129. Which data structure gives the minimum (or maximum) element in O(1) and inserts in O(log n)?

A) Heap (priority queue)
B) Linked list
C) Stack
D) Array

Q130. What is the output?

```
# Python
print(list(range(1, 10, 2)))
```

A) [0, 2, 4, 6, 8]
B) [2, 4, 6, 8, 10]
C) [1, 3, 5, 7, 9]
D) [1, 3, 5, 7]

Q131. What is the output of this C code?

```
// C
int i;
for (i = 0; i < 4; i++);
printf("%d", i);
```

A) 5
B) 0
C) 3
D) 4

Q132. What is the output of this C code?

```
// C
int a = 7, b = 3;
printf("%d", a / b);
```

A) Error
B) 2
C) 1
D) 3

Q133. What is the output of this C code?

```
// C
int arr[] = {5, 10, 15};
printf("%d", (int)(sizeof(arr) / sizeof(arr[0])));
```

A) 2
B) 3
C) 4
D) 1

Q134. What is the output?

```
# Python
print(list(zip([1, 2], ['a', 'b'])))
```

A) Error
B) [1, 2, 'a', 'b']
C) {1: 'a', 2: 'b'}
D) [(1, 'a'), (2, 'b')]

Q135. What is the output?

```
# Python
print(dict(zip([1, 2], ['a', 'b'])))
```

A) {1: 'a', 2: 'b'}
B) Error
C) [(1, 'a'), (2, 'b')]
D) {'a': 1, 'b': 2}

Q136. What is the output?

```
# Python
def bs(a, t):
    lo, hi = 0, len(a) - 1
    while lo <= hi:
        mid = (lo + hi) // 2
        if a[mid] == t:
            return mid
        elif a[mid] < t:
            lo = mid + 1
        else:
            hi = mid - 1
    return -1
print(bs([1, 3, 5, 7, 9], 7))
```

A) Error
B) 2
C) 3
D) 4

Q137. In a Java class, which access modifier makes a member visible only within the same class?

A) private
B) public
C) default (no modifier)
D) protected

Q138. What is the output?

```
# Python
a = [1, 2]
b = a
c = [1, 2]
print(a is b, a is c)
```

A) False True
B) True True
C) False False
D) True False

Q139. What is the output?

```
# Python
m = [[1, 2], [3, 4], [5, 6]]
print([x for row in m for x in row])
```

A) [[1, 2], [3, 4], [5, 6]]
B) [1, 2, 3, 4, 5, 6]
C) Error
D) [1, 3, 5]

Q140. What is the number of edges in a tree with n nodes?

A) n
B) 2n
C) n − 1
D) n + 1

Q141. What is the output?

```
# Python
print(bin(15).count('1'))
```

A) 3
B) Error
C) 5
D) 4

Q142. What is the output?

```
# Python
print(bin(10))
```

A) Error
B) 10
C) 0b1010
D) 1010

Q143. What is the output?

```
# Python
print(5 ^ 3)
```

A) 6
B) Error
C) 5
D) 3

Q144. What is the output?

```
# Python
print(1 << 3)
```

A) 16
B) 8
C) 2
D) 1

Q145. Which technique stores results of sub-problems to avoid recomputing them?

A) Backtracking only
B) Dynamic programming (memoisation)
C) Brute force
D) Greedy method

Q146. What is the output?

```
# Python
a, b = 1, 2
a, b = b, a
print(a, b)
```

A) 2 1
B) 1 1
C) 1 2
D) 2 2

Q147. What is the output?

```
# Python
def triple(f):
    def wrap():
        return f() * 3
    return wrap
@triple
def ten():
    return 10
print(ten())
```

A) 20
B) Error
C) 10
D) 30

Q148. What is the output?

```
# Python
a = [1, 2, 3]
a = a[-1:] + a[:-1]
print(a)
```

A) [1, 2, 3]
B) [2, 3, 1]
C) [3, 1, 2]
D) [3, 2, 1]

Q149. What is the output?

```
# Python
print(sorted(set([3, 1, 2, 1])))
```

A) [1, 1, 2, 3]
B) [1, 2, 3]
C) [3, 2, 1, 1]
D) {1, 2, 3}

Q150. What is the output?

```
# Python
print('Abc'.swapcase())
```

A) Abc
B) abc
C) aBC
D) ABC

PART B

Answer Key

BEGINNER Questions 1 – 50

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	B	2	A	3	C	4	D	5	A
6	B	7	C	8	D	9	B	10	A
11	D	12	C	13	A	14	C	15	B
16	D	17	A	18	D	19	C	20	B
21	D	22	C	23	A	24	B	25	B
26	C	27	A	28	D	29	A	30	C
31	D	32	B	33	A	34	B	35	D
36	C	37	B	38	A	39	C	40	D
41	D	42	B	43	C	44	A	45	D
46	C	47	B	48	A	49	A	50	D

INTERMEDIATE Questions 51 – 100

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
51	C	52	B	53	A	54	B	55	C
56	D	57	A	58	C	59	D	60	B
61	D	62	A	63	C	64	B	65	A
66	D	67	B	68	C	69	A	70	C
71	B	72	D	73	C	74	D	75	A
76	B	77	A	78	C	79	D	80	B
81	D	82	B	83	C	84	A	85	C
86	B	87	A	88	D	89	C	90	D
91	B	92	A	93	C	94	A	95	B
96	B	97	D	98	A	99	B	100	C

ADVANCED Questions 101 – 150

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
101	B	102	D	103	A	104	C	105	D
106	A	107	B	108	C	109	C	110	A
111	D	112	B	113	B	114	A	115	D
116	C	117	A	118	B	119	C	120	D
121	C	122	D	123	A	124	B	125	D
126	C	127	A	128	B	129	A	130	C

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
131	D	132	B	133	B	134	D	135	A
136	C	137	A	138	D	139	B	140	C
141	D	142	C	143	A	144	B	145	B
146	A	147	D	148	C	149	B	150	C

PART C

How to Solve — Step-by-Step Solutions

Each solution shows the correct option and a simple way to reach it. Try to understand the method, not just the answer — the same method works for many similar questions.

BEGINNER Questions 1 – 50

Q1 Answer: B) 8

The + operator adds two integers: 5 + 3 = 8.

Q2 Answer: A) 53

Both values are strings, so + joins (concatenates) them: '5' + '3' = '53'.

Q3 Answer: C) Stack

A stack is LIFO: the last item pushed is the first popped, like a pile of plates.

Q4 Answer: D) Queue

A queue is FIFO: the first item added is the first removed, like a ticket line.

Q5 Answer: A) 3

// is floor division in Python. 7 // 2 = 3.

Q6 Answer: B) 1

% gives the remainder. 7 ÷ 2 = 3 remainder 1.

Q7 Answer: C) 0 1 2

range(3) produces 0, 1, 2 — it stops before 3.

Q8 Answer: D) def

Python functions are defined with def, e.g. def greet():

Q9 Answer: B) 0

Indexing starts at 0 in Python, C, Java and most languages.

Q10 Answer: A) 10

x is 5, then x = x * 2 makes it 10.

Q11 Answer: D) total_marks

Variable names cannot start with a digit, contain spaces or contain '-'. Underscores are allowed.

Q12 Answer: C) Hello

printf prints the text inside quotes exactly: Hello.

Q13 Answer: A) True

10 is greater than 5, so the comparison is True.

Q14 Answer: C) A step-by-step procedure to solve a problem

An algorithm is a clear set of steps that leads to a solution.

Q15 Answer: B) 3

len() returns the number of items in a list. [10, 20, 30] has 3 items.

Q16 Answer: D) 20

nums[1] is the second element (index 1), which is 20.

Q17 Answer: A) 30

Negative index −1 means the last element: 30.

Q18 Answer: D) #

Python uses # for comments. C, C++ and Java use //.

Q19 Answer: C) Even

8 % 2 == 0 is True, so 'Even' is printed.

Q20 Answer: B) HELLO

upper() converts every letter to uppercase.

Q21 Answer: D) repeat-until

C has for, while and do-while. repeat-until is from Pascal.

Q22 Answer: C) 6

The loop adds 1 + 2 + 3 = 6.

Q23 Answer: A) Sends a value back to the caller and ends the function

return exits the function and gives back a result.

Q24 Answer: B) 9

** is the power operator in Python. 3 ** 2 = 3 × 3 = 9.

Q25 Answer: B) An error or flaw in a program

A bug is a mistake in code that causes wrong behaviour.

Q26 Answer: C) <class 'int'>

type(5) tells you the type of the value 5, which is an integer.

Q27 Answer: A) Boolean

A Boolean holds one of two values: true or false.

Q28 Answer: D) Hi Hi Hi

The loop runs 3 times and prints 'Hi ' each time.

Q29 Answer: A) 11

x++ increases x by 1. 10 + 1 = 11.

Q30 **Answer: C) ==**

== compares values. A single = assigns a value.

Q31 **Answer: D) [1, 2, 3, 4]**

append() adds the item to the end of the list.

Q32 **Answer: B) Python**

The string is stored in s and printed as it is.

Q33 **Answer: A) P**

s[0] is the first character: 'P'.

Q34 **Answer: B) input()**

input() reads a line typed by the user as a string.

Q35 **Answer: D) Object-Oriented Programming**

OOP organises code around objects that combine data and behaviour.

Q36 **Answer: C) A blueprint for creating objects**

A class defines the properties and methods; objects are created from it.

Q37 **Answer: B) 5**

max() returns the largest value: 5.

Q38 **Answer: A) 2.5**

/ is true division in Python 3: 5 / 2 = 2.5.

Q39 **Answer: C) Good**

score is 75, which is >= 50, so 'Good' is printed.

Q40 **Answer: D) abcabc**

Multiplying a string by a number repeats it.

Q41 **Answer: D) Finding and fixing errors in code**

Debugging is the process of locating and correcting bugs.

Q42 **Answer: B) C**

C code is compiled into machine code before running. HTML, CSS and Markdown are not programming languages that are compiled.

Q43 **Answer: C) 3**

min() returns the smallest value. In [3, 7, 5], the smallest is 3.

Q44 **Answer: A) 15**

The function returns a + b = 10 + 5 = 15.

Q45 **Answer: D) A named storage location for data**

A variable is a name that holds a value which can change.

Q46 **Answer: C) False**

'and' is True only if both sides are True. False makes the whole result False.

Q47 **Answer: B) True**

'or' is True if at least one side is True.

Q48 **Answer: A) [2, 3]**

Slicing a[1:3] takes index 1 up to (but not including) 3: [2, 3].

Q49 **Answer: A) Bubble sort**

Bubble sort compares adjacent items and swaps them, so large values 'bubble up' to the end.

Q50 **Answer: D) 4**

The while loop runs while i < 4 and adds 1 each time. It stops when i becomes 4.

INTERMEDIATE Questions 51 – 100

Q51 Answer: C) [1, 2, 3, 4]

b = a does not copy the list; both names refer to the same list. Appending through b changes a too.

Q52 Answer: B) O(log n)

Each step halves the search space, so it takes about $\log_2 n$ steps.

Q53 Answer: A) olleh

The slice [::-1] reverses a string.

Q54 Answer: B) 120

This is a recursive factorial: 5! = 5 × 4 × 3 × 2 × 1 = 120.

Q55 Answer: C) O(n^2)

Bubble sort has two nested loops over the data, giving O(n^2) in the worst case.

Q56 Answer: D) 10 20

The function receives copies of a and b (call by value), so swapping inside it does not change the originals.

Q57 Answer: A) Encapsulation

Encapsulation bundles data with methods and restricts direct access (e.g. private fields).

Q58 Answer: C) Inheritance

Inheritance allows a class to derive features from another class.

Q59 Answer: D) {'a': 1, 'b': 2}

This creates a dictionary with two key-value pairs and prints it.

Q60 Answer: B) 3

A set keeps only unique values. {1, 2, 2, 3} becomes {1, 2, 3}, so its length is 3.

Q61 Answer: D) 1 4 9

The loop prints the square of 1, 2 and 3.

Q62 Answer: A) Stack

Push each opening bracket; pop when a closing one appears. The stack's LIFO order matches nested brackets.

Q63 Answer: C) [0, 2, 4, 6, 8]

List comprehension doubles each number in range(5): 0, 2, 4, 6, 8.

Q64 Answer: B) 15

Both are int values, so + adds them: 10 + 5 = 15.

Q65 Answer: A) Sum: 105

Java evaluates left to right. "Sum: " + 10 becomes a string "Sum: 10", then + 5 appends '5' → "Sum: 105".

Q66 Answer: D) Hash table

Hash tables compute the position directly from the key, giving constant average lookup time.

Q67 Answer: B) 2

count() returns how many times a value appears. The word 'level' has 'l' at the start and the end → 2.

Q68 Answer: C) 5

find() returns the index of the first match. Counting from 0: h=0, e=1, l=2, l=3, o=4, space=5. So the space is at index 5.

Q69 Answer: A) The same method name behaving differently for different objects

Polymorphism ('many forms') lets one interface or method name work in different ways, e.g. area() for Circle and Square.

Q70 Answer: C) 3

The loop runs for i = 0, 1, 2 — three times — and count increases each time.

Q71 Answer: B) 5

x++ is post-increment: the current value (5) is used first, then x becomes 6.

Q72 Answer: D) 6

++x is pre-increment: x becomes 6 first, then that value is assigned.

Q73 Answer: C) In-order

In-order (left, root, right) visits BST nodes in ascending order.

Q74 Answer: D) [3, 2, 1]

reverse() reverses the list in place; printing the list then shows [3, 2, 1].

Q75 Answer: A) None

list.reverse() changes the list in place and returns None. Printing its return value shows None.

Q76 Answer: B) A sequence of nodes where each node points to the next

Each node stores data and a reference (pointer) to the next node.

Q77 Answer: A) 10

Assigning x = 5 inside the function creates a new local variable. The global x is not changed, so it stays 10.

Q78 Answer: C) 1 2 Fizz 4 Buzz

Multiples of 3 print 'Fizz' and multiples of 5 print 'Buzz'. For 1 to 5: 1, 2, Fizz, 4, Buzz.

Q79 Answer: D) 15

sum() adds all items: 1 + 2 + 3 + 4 + 5 = 15.

Q80 Answer: B) ['a', 'b', 'c']

list() on a string splits it into individual characters.

Q81 **Answer: D) ['red', 'green', 'blue']**

split(',') breaks the string at each comma into a list.

Q82 **Answer: B) Merge sort**

Merge sort splits the list, sorts each half and merges them, always in O(n log n).

Q83 **Answer: C) 2**

The loop stops (break) as soon as i becomes 2. The variable i keeps its last value, so print(i) after the loop shows 2.

Q84 **Answer: A) 0 1 3 4**

continue skips the rest of the loop body for i = 2, so 2 is not printed.

Q85 **Answer: C) (1, 2, 3)**

Parentheses with commas create a tuple, which prints with round brackets.

Q86 **Answer: B) TypeError — tuples cannot be changed**

Tuples are immutable, so assigning to an index raises a TypeError.

Q87 **Answer: A) new**

Objects are created with new, e.g. Car c = new Car();

Q88 **Answer: D) True**

'in' checks membership. 3 is in the list.

Q89 **Answer: C) $O(n^2)$**

An n × n matrix has $n \times n = n^2$ cells.

Q90 **Answer: D) 4**

The function doubles its input; it is called twice: double(double(1)) = double(2) = 4.

Q91 **Answer: B) 12**

int() converts the strings to numbers 3 and 4, and * multiplies them: 3 × 4 = 12.

Q92 **Answer: A) A special method that runs when an object is created**

The constructor (e.g. __init__ in Python, same-name method in Java) sets up a new object.

Q93 **Answer: C) Hello, Asha**

An f-string replaces {name} with the variable's value.

Q94 **Answer: A) 5**

Each loop removes the last digit (n //= 10) and adds 1 to count. 12345 → 1234 → 123 → 12 → 1 → 0: five steps, so it counts the digits = 5.

Q95 **Answer: B) [1, 2, 3, 4, 5]**

sorted() returns a new list in ascending order.

Q96 **Answer: B) Enqueue adds at the rear; dequeue removes from the front**

A queue adds at the back and removes from the front (FIFO).

Q97 **Answer: D) 10**

The function returns the default value 10 when no argument is passed.

Q98 **Answer: A) Exits the loop immediately**

break stops the loop completely. continue skips only the current iteration.

Q99 **Answer: B) abc**

lower() converts all letters to lowercase.

Q100 **Answer: C) 3**

index() returns the position of the first matching item. In [10, 20, 30, 40], 40 is at index 3.

ADVANCED Questions 101 – 150

Q101 Answer: B) [1] [1, 1]

The default list is created only once, when the function is defined. Each call appends to the same list.

Q102 Answer: D) 8

This prints the 6th Fibonacci number (fib(0)=0, fib(1)=1): 0, 1, 1, 2, 3, 5, 8 → fib(6) = 8.

Q103 Answer: A) O(2ⁿ)

Each call makes two more calls, so the call tree grows roughly as 2^n.

Q104 Answer: C) [[1, 0], [1, 0]]

[[0] * 2] * 2 copies the same inner list twice, so changing one row changes both.

Q105 Answer: D) 15

The function receives the address of a and adds 5 to the value at that address (call by reference using a pointer).

Q106 Answer: A) O(n²)

If the pivot is always the smallest or largest element (e.g. already sorted data with a poor pivot choice), quicksort degrades to $O(n^2)$.

Q107 Answer: B) 3

A Python list works as a stack. pop() removes and returns the last item pushed, which is 3 (LIFO).

Q108 Answer: C) 1

deque.popleft() removes from the front (FIFO). The first item added was 1.

Q109 Answer: C) Queue

BFS explores neighbours level by level, which needs FIFO order — a queue. DFS uses a stack (or recursion).

Q110 Answer: A) {'a': 2, 'b': 1}

The loop counts each character. 'a' appears twice and 'b' once; dicts keep insertion order.

Q111 Answer: D) 6

The nested loops run 2 × 3 = 6 times in total.

Q112 Answer: B) O(n²)

The outer loop runs n times and the inner loop runs n times for each, giving $n \times n = n^2$ operations.

Q113 Answer: B) O(log n)

i doubles each step (1, 2, 4, 8 ...), so the loop runs about $\log_2 n$ times.

Q114 Answer: A) false true

== compares object references; the two new String objects are different objects → false. equals() compares content → true.

Q115 **Answer: D) [2, 4]**

filter keeps only items where the lambda returns True (even numbers).

Q116 **Answer: C) [1, 4, 9]**

map applies the lambda (square) to each item.

Q117 **Answer: A) 10**

It adds the digits: 1 + 2 + 3 + 4 = 10.

Q118 **Answer: B) True**

The function checks if a string reads the same backwards. 'madam' reversed is 'madam'.

Q119 **Answer: C) Merge sort**

Merge sort keeps equal elements in their original order (stable) and is O(n log n). Standard quicksort and heapsort are not stable.

Q120 **Answer: D) 2**

The class counts how many objects were created using a class variable. Two objects are made.

Q121 **Answer: C) Woof**

Dog overrides the speak() method, so calling it on a Dog object uses the Dog version (polymorphism / method overriding).

Q122 **Answer: D) Error: division by zero**

Dividing by zero raises ZeroDivisionError, which the except block catches and handles.

Q123 **Answer: A) A C**

No exception occurs, so 'B' (the except block) is skipped. finally always runs.

Q124 **Answer: B) 10**

2^{10} = 1,024 ≥ 1,000, so at most 10 comparisons are needed.

Q125 **Answer: D) Dijkstra's shortest path algorithm**

Dijkstra always picks the nearest unvisited node next — a greedy choice that is optimal for non-negative weights.

Q126 **Answer: C) [1, 2, 3] [1, 2, 3, 4]**

a.copy() creates a separate list, so appending to b does not change a.

Q127 **Answer: A) 0 1 2**

A generator yields values one at a time. range(3) gives 0, 1, 2.

Q128 **Answer: B) 15**

reduce applies the function cumulatively: ((((1 + 2) + 3) + 4) + 5) = 15.

Q129 **Answer: A) Heap (priority queue)**

A binary heap keeps the smallest (min-heap) at the root and restores order in O(log n) after insertion.

Q130 **Answer: C) [1, 3, 5, 7, 9]**

range(1, 10, 2) starts at 1 and steps by 2 up to (not including) 10.

Q131 **Answer: D) 4**

The loop runs while i < 4. It stops when i becomes 4, then prints 4.

Q132 **Answer: B) 2**

Integer division in C drops the fraction: 7 / 3 = 2.

Q133 **Answer: B) 3**

The array has 3 elements; sizeof(arr) / sizeof(arr[0]) gives the element count.

Q134 **Answer: D) [(1, 'a'), (2, 'b')]**

zip pairs up items from both lists position by position.

Q135 **Answer: A) {1: 'a', 2: 'b'}**

dict(zip(...)) builds a dictionary using the first list as keys.

Q136 **Answer: C) 3**

This is binary search returning the index of 7 in the sorted list [1, 3, 5, 7, 9]. 7 is at index 3.

Q137 **Answer: A) private**

private restricts access to the declaring class. protected allows subclasses; public allows all.

Q138 **Answer: D) True False**

is checks identity. b = a points to the same list (True). c is a new list with equal content but a different object (False).

Q139 **Answer: B) [1, 2, 3, 4, 5, 6]**

The nested comprehension flattens the list of lists into one list.

Q140 **Answer: C) n − 1**

A tree is connected with no cycles; it always has exactly n − 1 edges.

Q141 **Answer: D) 4**

This counts the 1-bits in the binary form of 15 (1111) → 4.

Q142 **Answer: C) 0b1010**

bin() returns the binary string with the prefix '0b'.

Q143 **Answer: A) 6**

^ is bitwise XOR: a bit is 1 when the two bits differ. 5 = 101 and 3 = 011 → 110 = 6.

Q144 **Answer: B) 8**

<< shifts bits left. $1 << 3 = 1 \times 2^3 = 8$.

Q145 **Answer: B) Dynamic programming (memoisation)**

Dynamic programming saves answers to smaller sub-problems and reuses them, e.g. Fibonacci with memoisation runs in O(n).

Q146 **Answer: A) 2 1**

Python's tuple unpacking swaps values without a temporary variable.

Q147 **Answer: D) 30**

The decorator wraps the function and multiplies its result by 3: 10 × 3 = 30.

Q148 **Answer: C) [3, 1, 2]**

This rotates the list right by one: the last element moves to the front.

Q149 **Answer: B) [1, 2, 3]**

set() removes duplicates and sorted() returns them in ascending order as a list.

Q150 **Answer: C) aBC**

swapcase() turns uppercase letters into lowercase and lowercase into uppercase. 'Abc' becomes 'aBC'.